PLANNED RELOCATION IN ESPARZA COSTA RICA

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Disclaimer

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Abbreviations

Name	Abbreviation	Short Description
Community Development Associations	**ADCs**	Community-based organizations elected democratically
Federated College of Engineers and Architects	**CFIA**	Ensures excellence in the professional practice of engineering and architecture in Costa Rica
Housing Family Bond	**BFV**	Government housing subsidy that helps low-income families buy, build, or improve a home.
Housing Mortgage Bank	**BANHVI**	Public bank that finances Costa Rica's social housing system and administers the BFV.
International Organization for Migrations	**OIM**	United Nations organization working on migrations
Ministry of Environment and Energy	**MINAE**	Lead national institution in environmental policy.
Mixed Institute for Social Assistance	**IMAS**	Social protection agency providing financial aid to low-income households, including emergency subsidies during disasters.
Ministry of Housing and Human Settlements	**MIVAH**	Lead national institution in housing. Regulates and dictates policy and coordinates the social housing system.
National Risk Management System	**SNGR**	National coordination framework that integrates institutions managing disaster risk across all stages.
National Commission for Risk Prevention and Emergency Response	**CNE**	Lead national institution for disaster risk management, coordinating prevention, response, and recovery actions.
National Housing and Urbanism Institute	**INVU**	Institution responsible for urban planning, land development, and public housing projects.

Executive Summary

Executive Summary

This report argues that Whole Community Relocation is the strongest policy option to protect 96 households in Esparza, Costa Rica, from recurrent flooding and coastal erosion. Paired with a Transfer of Development Rights pilot framework, this approach offers a promising strategy for future risk-related relocations in the canton.

In the communities of La Plaza, La Cueva, and Caldera, 96 households face repeated flooding and coastal erosion that have caused permanent displacement, housing destruction, and loss of livelihoods. Current government responses, including emergency shelters, temporary cash transfers, and housing subsidies, were not designed for displacement and have consistently fallen short.

Against four criteria, risk reduction potential, cost, implementation feasibility, and social acceptance, Whole Community Relocation outperforms the other three alternatives. It is the only option that reaches all 96 households, preserves community cohesion, and has a realistic financing path through the Green Climate Fund. Government buyouts and land swaps are faster, but they only reach between 30% and 50% of households, fragment communities, and face significant legal and logistical barriers.

Whole Community Relocation carries an estimated cost of 1,440 million colones, or about 15 million per household, spread over five years. The implementation plan runs six phases over 5.5 to 6 years, starting with institutional coordination before community engagement, a 75% household participation threshold, and a contingency financing strategy if international funds fall short.

That said, Transfer of Development Rights shows promise as a market-driven solution for future relocations. It has no precedent in Costa Rica and faces significant uncertainty due to its dependency on private sector response, which makes a pilot framework the right next step rather than a full rollout.

Managed relocation of climate-exposed communities is one of the most complex challenges in disaster risk policy. The findings of this report point to Whole Community Relocation as the path that best balances risk reduction, feasibility, and community wellbeing for Esparza's most vulnerable families, and TDRs as a potential long-term solution for Costa Rica's growing displacement challenges.

Introduction

Climate change is one of the main challenges on today's public agenda. Changes in climate patterns, the intensification of atmospheric events, and sea-level rise, combined with high vulnerability and exposure, create significant risks to human life in certain areas (IPCC, 2023).

According to recent projections, Costa Rica could experience an increase in average annual temperature between 2.5°C and 3.5°C by 2080 (IPCC, 2014), along with rainfall increases of up to 50%, and up to 90% in coastal districts (MINAE, 2022). Sea levels are expected to rise by 0.30 meters, with tides reaching up to 4.58 meters, which may affect coastal areas, particularly along the Pacific coast (MINAE, 2022; IPCC, 2014). These trends point to an increase in extreme climate events that, coupled with existing vulnerability, could intensify the number and severity of disasters.

Between 2005 and 2017, Costa Rica recorded roughly $2.210 billion in infrastructure losses (MINAE, 2018). From 2021 to 2025, an additional $1.604 billion in damages were registered, of which $64 million (4%) corresponded to housing infrastructure (CNE, 2025). However, these figures reflect only events declared national emergencies and exclude localized but frequent disasters like coastal flooding. They also omit indirect costs related to production disruptions, school closures, and ecosystem impacts. Projections indicate that annual losses could exceed $7 billion by 2030 and $30 billion by 2050 (MINAE, 2018).

Housing-related displacement is one of the most pressing consequences. In 2017, Costa Rica recorded 10,678 people displaced due to floods and storms (OIM, 2022). Since 2021, national-scale events have forced at least 8,452 people (equivalent to 2,119 households) to leave their homes temporarily or permanently, and more than 50% require relocation because their homes are in high-risk areas (CNE, 2025). This situation is expected to worsen. The Pacific Coast, already affected by coastal erosion, may experience increased internal migration toward urban centers (OIM, 2022), particularly in highly exposed and socioeconomically vulnerable territories such as Esparza.

This report examines these challenges through the lens of disaster risk management, which analyzes the components of disaster risk (hazard, exposure, and vulnerability) and all stages from prevention to recovery and resettlement (UNGA, 2016; Sandoval et al., 2023; White, 2022; IPCC, 2023; United Nations, 2015). When prevention and mitigation alone cannot address the level of risk, planned relocation emerges as a necessary alternative. This report focuses on communities in Esparza where that threshold has been reached. It evaluates four alternatives for planned relocation, draws on original cost estimates and evidence from comparable cases, and recommends Whole Community Relocation as the most effective and socially legitimate option, with a six-phase implementation plan for the CNE.

Problem statement

In the canton of Esparza, residents from La Plaza, La Cueva, and Caldera communities face high vulnerability to hydrometeorological hazards (Calvo et al, 2024; E. Aguilar, personal communication, October 14, 2025), leading to recurrent impacts on their homes, health, and livelihoods. In the absence of safe and sustainable relocation solutions, many displaced families are forced to return to high-risk areas. This situation reflects an insufficient institutional capacity to provide planned relocation alternatives in disaster contexts, perpetuating a cycle of vulnerability and exposure to future extreme events.

Client Overview

The National Commission for Risk Prevention and Emergency Response (CNE) is the lead institution in charge of risk management and coordinating emergency response in Costa Rica. Created in 1969 and reporting directly to the President, it operates under Law 8488, which establishes the National Risk Management System (SNGR) and defines the formal mechanisms for interinstitutional coordination. This system includes sectoral and institutional committees that coordinate preparedness, response, and recovery actions for events such as floods, landslides, and storm surges (CNE, 2026).

The intensification of climate events has increased pressure on this system, especially in communities exposed to river and coastal flooding. When families displaced by disasters are forced to return to high-risk areas due to the lack of planned relocation alternatives, the capacity of the CNE and the System to fulfill their mandate of prevention and mitigation is weakened, creating chronic risks to people’s lives and livelihoods. For this reason, addressing this issue properly is essential to protect vulnerable populations and find solutions before changes in climate patterns deepen the impacts on these communities.

Background

a. *Geographic and socioeconomic context: Esparza*

The canton of Esparza, located in the Central Pacific region of Costa Rica, borders the Gulf of Nicoya and the city of Puntarenas. With an area of 217.23 km^2, its territory includes mountainous areas in the northeast and coastal plains in the southwest, crossed by rivers such as the Jesús María and the Barranca. These rivers are essential for agricultural and fishing activities, but they also pose hydrometeorological risks due to their potential to overflow (Municipalidad de Esparza, 2025).

The canton has an estimated population of 44,699 inhabitants and includes communities with high levels of poverty, even though it is classified as having medium-high human development (PNUD, 2024; Calvo et al., 2024). Its economy is based on commerce, tourism, agriculture, livestock, and fishing, although it faces structural limitations related to formal employment, infrastructure, and land-use planning (Calvo et al., 2024).

b. *Exposure, natural threats, and vulnerabilities in Esparza*

Since 1935, eighteen emergencies have been recorded in the canton, eight of them after 2016, reflecting a rise in extreme events. The main natural hazards are river and rain-induced flooding, landslides on unstable slopes, and coastal flooding driven by wave action and sea-level rise (Municipalidad de Esparza, 2025).

The district of Caldera is one of the most affected areas in Esparza. It has 3,254 inhabitants, and at least 1,000 families – over 30% of the population – face a direct risk of displacement (Municipalidad de Esparza, 2025; Calvo et al., 2024). The communities of La Plaza, La Cueva, and Caldera show the highest social, economic, and climate vulnerability. Wave events in 2014 and 2015 caused total housing losses and repeated damage to tourism and port infrastructure.

Coastal erosion and sea-level rise have led to eviction orders and the revocation of land-use permits in areas such as La Plaza, declared uninhabitable by the CNE. Homes in these zones have been buried under 1.5 to 2 meters of sand, with severe damage to sanitation systems, and some remain informal settlements where residents do not own the land (Calvo et al., 2024).

These conditions have caused both temporary and permanent displacement. Many families have had to leave their homes due to frequent and intense wave events, leading to temporary relocations or stays with relatives within the canton (Calvo et al., 2024). Although ineffective, there have been institutional efforts to address this issue, which are described in the following section.

c. *Previous policies and institutional barriers*

Costa Rica has no mechanism specifically designed to address disaster-induced displacement (Navas, 2025). In practice, authorities have used existing social protection and housing programs, which fall into two categories: temporary assistance during emergencies and permanent housing mechanisms for low-income families.

During an emergency, the CNE and local municipal emergency committees activate temporary shelters for families who must immediately evacuate. In parallel, the Mixed Institute for Social Assistance (IMAS) provides a short-term cash transfer for up to three months to help affected households cover basic needs such as rent, food, and transportation (Calvo et al., 2024; Centeno-Morales et al., 2020). These mechanisms address immediate safety needs but are not designed to offer permanent housing solutions.

Once the emergency phase ends, families whose homes are declared severely damaged or uninhabitable may apply for the Housing Family Bond (BFV), administered by the Housing Mortgage Bank (BANHVI). The BFV is a one-time state subsidy that helps low- and middle-income families purchase land, build, or buy a home (BANHVI, n.d). Its maximum amount is approximately 9,300,000 colones, covering roughly 35% of a home's market value (Navas, 2025). Families must finance the remainder through savings or debt, which places a significant burden on households that have just lost assets and income. The application process can take up to five years to complete, and families just over the poverty threshold are excluded.

Families in extreme poverty may also access Social Interest Housing Projects, coordinated by the National Housing and Urbanism Institute (INVU) with the support of BANHVI. In these projects, affected families are grouped to relocate them collectively to new, basic-standard developments, generally covering land and construction costs for eligible households (INVU, 2025). Past experiences in Calle Lajas, Barrio Fátima, Cinchona, and Esparza have shown mixed results, with recurring barriers related to poor inter-institutional coordination, the absence of a specific emergency housing mechanism, and limited resource availability (OIM, 2022; Centeno-Morales et al., 2020; Fernández Arce et al., 2020). Like the BFV, processing times can extend to five years, and it excludes households over the extreme poverty line.

The CNE does not have direct participation in these permanent housing programs. As the institution responsible for emergency response and disaster risk reduction, its role is currently focused on the emergency phase. Long-term housing provision falls under the Ministry of Housing (MIVAH) and BANHVI, which operate under different mandates, timelines, and criteria. This means that when permanent displacement occurs, no coordinated mechanism exists that combines emergency urgency with housing capacity. The 2012 relocation attempt in San Martín, Esparza illustrates this gap: eligible families were assigned to a Social Interest Housing Project located 130 km from the community, the process excluded community participation, and most families chose not to relocate. Many remain in high-risk areas today (Calvo et al., 2024; Polack et al., 2025).

This situation makes clear that the communities in La Plaza, La Cueva, and Caldera require a different approach, one designed specifically around planned relocation. The next section evaluates four alternatives that respond to this need, examining their potential for risk reduction, cost, implementation feasibility, and social acceptance.

Figure 1. Coastal flooding in Caldera, 2014

Source: La Nación, 2014. Floodings in 2014 affected 70 families in total. Waves up to 3.5 meters hit the coast. Municipal authorities advised the communities about the risks of staying and offered temporary help to evacuate the area

Alternatives,

Alternatives

When prevention and mitigation options are exhausted, planned relocation or managed retreat becomes necessary (Brookings Institution, 2015; World Bank, 2018; GCC, 2020). This section details and evaluates four alternatives in this category for the target communities in Esparza against the criteria in *Table 1*.

Table 1. Evaluation Criteria

1. Potential Risk Reduction Number of households that could be relocated out of high-risk areas and the expected timeline to achieve it. Coverage depends on whether the alternative conditions are based on formal property title or allow broader inclusion.	**2. Cost** Total and per-household cost estimated using fiscal property values, real estate market data, and the costing tool from Polack et al. (2025). Costs include government expenditures such as subsidies, construction, administration, and site restoration.
3. Implementation Feasibility Administrative and legal viability of each alternative, assessed as High, Medium, or Low based on the existence of legal frameworks, institutional precedent, and the level of coordination required among public actors.	**4. Social Acceptance** Expected level of community support based on comparable cases, assessed qualitatively. Considers effects on community cohesion, access to livelihoods, and the degree of meaningful participation in the design and execution of the relocation.

Alternative A: Government buyouts with leaseback variant

This alternative groups two modalities under a single government acquisition mechanism. In the direct buyout, the government purchases homes in high-risk areas, families vacate at the time of purchase, and structures are demolished with land restored to prevent future development. In the leaseback variant, the government purchases the home but the former owner remains as a temporary occupant without paying rent, allowing for a gradual transition to permanent housing; demolition and land restoration follow once the family relocates (White, 2022). Both modalities share the same core mechanism and final outcome; the leaseback modifies only the implementation timeline and adds an occupancy management layer.

Buyouts are most commonly implemented as voluntary, parcel-by-parcel transactions, which allow for gradual implementation but often result in fragmented acquisitions that limit risk reduction and land-use benefits (GCC, 2020). Neighborhood-scale buyouts, as seen in Isle de Jean Charles and Christchurch, New Zealand, target multiple properties at once and enable land conversion to open space (UNHCR, 2017; Ajibade et al., 2022). The leaseback variant has been implemented through revolving loan mechanisms, where rental payments help repay the initial purchase cost before demolition, offering households flexibility to plan their relocation and preventing resale of homes in hazardous areas (GCC, 2020; White, 2022; Keeler et al., 2022).

Evidence on buyouts is mixed. Some studies show benefits such as complete risk elimination and greater household autonomy (Stern, 2022; Ajibade et al., 2022). However, programs may leave communities unprotected if purchase prices are set below market values (Miao and Davlasheridze, 2022) and can exacerbate inequities in low-income communities through long, costly processes (Siders, 2019). Most findings come from the United States, limiting direct generalizability to Costa Rica's institutional context.

In Esparza, households ready to relocate would immediately follow the direct buyout track, while those needing a transition would use the leaseback variant. The Municipality identifies and prioritizes at-risk properties, the CNE verifies eligibility and transfers funds, and acquisitions are voluntary. Key risks include low participation, exclusion of households without formal property titles, and prolonged temporary occupation in high-risk areas if permanent relocation is delayed.

Potential risk reduction.

Under the medium coverage scenario, this alternative would reach 29 of the 96 households (30%), since eligibility requires a formal property title. This excludes roughly 70% of households, corresponding to the most vulnerable families, many located in informal settlements or the maritime-terrestrial zone (Polack et al., 2025; Calvo et al., 2024). In terms of timing, the direct buyout can be completed in approximately one year, since it operates under existing legal frameworks and requires no new infrastructure. However, for households that need a leaseback, effective risk reduction is delayed by five to seven years, the maximum period of temporary occupation.

Cost.

Cost estimates use Ministry of Finance guidelines and the costing tool from Polack et al. (2025). A critical cross-cutting finding is the gap between fiscal and market values: the average fiscal value of a house is approximately ₡10.4 million per household, while the most affordable home in the formal market costs around ₡37 million, meaning families compensated at fiscal value face a severe housing shortfall.

For the 29 eligible households, the total aggregate cost is estimated at ₡449 million. The cost per household is ₡14.1 million for direct buyout and ₡15.5 million with leaseback, including valuation studies, payment of fiscal value, legal and registration costs, and site recovery. Even the maximum BFV subsidy is not enough to close the gap between fiscal value and market price, leaving families with a deficit of more than ₡26 million to purchase the most affordable home available.

Implementation feasibility.

The direct buyout has the highest feasibility among all alternatives. It operates under the National Risk Management System (Law 8488) and existing expropriation mechanisms (Law 7495), with no new legislation required. However, the leaseback variant has low feasibility: it requires the State or Municipality to act as landlord, which is not contemplated in current legal frameworks and has no precedent in disaster contexts in Costa Rica.

Social acceptance.

Both modalities present low social acceptance, as they operate individually and fragment the community. The fiscal-to-market value gap suggests that families would be forced to move away from their livelihoods, replicating the dynamic that caused the failure of San Martín (Calvo et al., 2024). While the leaseback allows for temporary proximity, the end result is still community dispersion. Participation is limited to a basic consultation level: the purchase is voluntary, but the community is not involved in the design of the solution.

Alternative B: Whole Community Relocation (WCR)

Whole community relocation involves moving an entire community together to a safer location, with the goal of reducing disaster risk while preserving social, cultural, and economic ties. Unlike parcel-by-parcel buyouts, this approach avoids neighborhood fragmentation and prioritizes collective decision-making, social cohesion, and cultural continuity, which is especially relevant for communities with strong territorial identities (White, 2022).

International experience shows mixed results. Large-scale relocations in Colombia faced delays and low completion rates, while Argentina's assisted self-build programs promoted community participation but required longer timelines (World Bank, 2018; Pérez & Zelmeister, 2011). Cases such as Vunidogoloa in Fiji show that strong public subsidies and local leadership can support successful integration of housing, infrastructure, and livelihoods, while experiences in Indonesia confirm that livelihood losses remain a key risk even when physical exposure is reduced (Tronquet, 2015; Balachandran et al., 2022).

WCR shows positive results when large numbers of people must be relocated together: key success factors include community participation and access to new livelihoods (Ajibade et al., 2022; Pinter, 2021; Koslov, 2016). However, challenges include high costs, procedural barriers, and livelihood disruption, since relocated residents can experience income losses in the years following relocation (Hoang & Noy, 2023; Polack et al., 2025; Pinter, 2021). As the San Martín case illustrates, these initiatives must consider access to livelihoods, jobs, and services, and must involve communities to align with their identity and ways of life (Calvo et al., 2024; Polack et al., 2025).

In Esparza, this alternative follows a participatory, project-based model studied by IOM (Calvo et al., 2024; Polack et al., 2025). The process begins with community engagement to assess collective willingness to relocate and define minimum acceptable conditions, led by the CNE and the Municipality. A safe relocation site is then identified and acquired, and the new settlement is designed through participatory planning integrating housing, livelihoods, and services.

The CNE leads construction and coordinates phased relocation with social support. The original site is later cleared through demolition and environmental restoration. Financing comes from international resources managed through the National Emergency Fund. Key risks include livelihood losses for households tied to the original location and the high administrative complexity of implementation.

Potential risk reduction.

WCR would cover 100% of the 96 affected households, since eligibility does not depend on formal property title. Implementation is estimated at approximately five years, based on prior experiences (OIM, 2022; Pinter, 2021; Calvo et al., 2024), given its multiple phases: community participation, planning, land acquisition, construction, and relocation.

Cost.

WCR is the highest-cost alternative in aggregate: ₡1,440 million in total, equivalent to ₡15 million per household, based on the Polack et al. (2025) costing tool for Caldera. This per-household figure is significantly lower than the market price of a home (₡37 million minimum) because it covers only social interest housing, modest constructions of approximately 45 to 60 m^2 with basic materials, on land assumed to be subsidized or donated by the State, and without a seller's profit margin.

The aggregate cost should be put in context: between 2021 and 2025, Costa Rica recorded ₡1,604 million in disaster damages (CNE, 2025), and annual losses could exceed ₡7,000 million by 2030 (MINAE, 2018). The WCR investment is a fraction of current losses and would be distributed over five years (approximately ₡288 million per year). As proposed by the CNE, financing would come from international sources, including climate funds and bilateral cooperation.

Implementation feasibility.

WCR has a medium implementation feasibility, as it can operate under Law 8488 without new legislation, but it requires strong inter-institutional coordination among the CNE, INVU, BANHVI, IMAS, and the municipality, among others. The main barrier is governance fragmentation and the absence of a dedicated relocation agency. Domestic evidence is mixed: San Martín is a negative precedent, while Cinchona, Calle Lajas, and Barrio Fatima show more favorable results with varying degrees of success (OIM, 2022; Centeno-Morales et al., 2020; Fernandez Arce et al., 2020).

Social acceptance.

WCR is the only alternative explicitly designed to preserve social, cultural, and economic ties through the collective relocation of the entire community, which is why its social acceptance is classified as high.

The model includes infrastructure for productive activities and is designed for a nearby site in Esparza, according to Polack et al. (2025). It incorporates free, prior, and informed consent (FPIC) processes and participatory planning in the form of community workshops, which the international literature identifies as the factors with the greatest influence on the success of planned relocations (Ajibade et al., 2022; Pinter, 2021).

Participation would be structured through community assemblies, the formation of a relocation committee, and household-level consultations, following the model documented by Calvo et al. (2024) and the FPIC framework.

Alternative C: Land swaps

Land swaps involve the permanent exchange of land between two or more owners, often with additional financial support. They are used to move households from high-risk areas to safer locations when public land is available (GCC, 2020).

International evidence is limited to descriptive case studies with no rigorous evaluations. In Guatemala, post-disaster relocations failed due to lack of basic services, and in Vietnam, land reassignment combined with subsidized loans led to high household debt and livelihood losses, a pattern directly relevant to Esparza (World Bank, 2018; UNHCR, 2017).

In Esparza, the CNE would identify nearby state-owned land (El Roble de Puntarenas, Esparza downtown, or La Trinidad de Orotina; see Figure 2) for potential donation, requiring legislative approval (Asamblea Legislativa, 1949). Voluntary swap agreements would be signed between the State and eligible families, with income-based housing subsidies from the CNE. Key risks include the exclusion of households without formal land ownership, limited nearby land availability, community fragmentation, and legal delays.

Figure 2. Potential Land Swaps Locations

Source: Google Maps. Distances from current location are 9.7 Kms for Site 1, 15 Kms for Site 2, and 11.3 Kms for site 3.

Potential risk reduction.

Under the high scenario, this alternative could reach up to 48 households (50%), since land donations could extend coverage beyond what formal titling allows, though these processes are politically complex and uncommon. Implementation could take up to four years, depending mainly on legislative approval.

Cost.

The total estimated cost is ₡818 million, equivalent to ₡17 million per household, the highest per-capita cost among the alternatives. This reflects the fiscal value of the donated land, estimated using the Map of Homogeneous Zones of El Roble, Puntarenas (Ministerio de Hacienda, 2018), selected for its similar characteristics to Caldera in terms of proximity to the coast, transportation, and public services. The amount includes family identification, land donation or exchange, construction subsidies, and recovery of the original site.

Implementation feasibility.

In this case, feasibility is low. The transfer of public land requires legislative approval (Asamblea Legislativa, 1949 – *art 141*), which introduces political uncertainty and long timelines. Also, there are no precedents for land swaps for disaster relocation in Costa Rica.

Social acceptance.

Community cohesion depends on whether families relocate together or dispersed; individual construction introduces inequality risks. Livelihoods depend on the location of the assigned land, which could move some families away from their current economic activities (tourism, fishing, port activities). Like buyouts, this alternative maintains a basic participation level where people voluntarily join the program but are not involved in its design, which is why its social acceptance is considered medium.

Alternative D: Transfer of Development Rights (TDRs)

TDRs use market incentives to redirect development from risk areas to safer areas with higher building capacity (GCC, 2020). Through zoning rules, the government allows property owners in sending areas to transfer development rights to developers in receiving areas, who can then build at a higher density. In the Esparza context, the model combines regulatory incentives for developers (density and height bonuses, municipal tax exemptions) with public housing subsidies and a mandatory requirement that at least 10% of units be designated for emergency housing.

There is no empirical evidence on TDRs specifically for disaster-induced relocation. Related market-based approaches show that outcomes depend heavily on local real estate conditions and regulatory capacity (GCC, 2020), and that developer subsidies can risk gentrification and displacement of low-income households if not carefully managed (Balachandran et al., 2022; World Bank, 2018). These limitations apply directly to Esparza's smaller, less formalized property market.

In Esparza, the TDR model transfers development rights from high-risk zones to safe receiving areas through a Municipal program, with regulatory incentives (density bonuses, tax exemptions) and a requirement that at least 10% of units be designated for emergency housing. Financing follows a public-private cost-sharing scheme with differentiated subsidies according to income status. Key risks include political resistance to higher density, enforcement challenges, and social fragmentation if affordable units are not well integrated.

Potential risk reduction.

TDRs do not condition access on property title, giving them a potential coverage of 100% (96 households). However, effective coverage depends on private sector participation, and since there are no prior experiences with this alternative in the country, realistic coverage is uncertain. Conservatively, and considering the dynamism of the real estate market in the western region, where cantons like Orotina and Esparza show high growth in new residential construction relative to unoccupied housing (CFIA, 2023), a realistic coverage of approximately 75% (72 households) is estimated.

The regulatory framework could be established in 1 to 2 years, but the effective relocation of households depends on when developers build and assign units, which could extend the timeline to 3 to 4 years or more, with significant uncertainty.

Cost.

The total estimated cost is ₡1,073 million, equivalent to ₡14.9 million per household, including technical studies, tax exemptions, market-value-based construction subsidies, relocation assistance, and site recovery. A significant portion of these costs are foregone tax revenues, not direct budget expenditures. The direct budget portion, mainly construction subsidies, amounts to approximately ₡868 million, considerably less than the direct cost of WCR.

Implementation feasibility.

TDRs require a new municipal regulatory framework, which involves approval by the Municipal Council and modifications to local land use plans. They also depend on the response of private developers in a relatively small local market. There are no precedents in Costa Rica for the use of TDRs in disaster relocation contexts, reason why the implementation feasibility is limited.

Social acceptance.

Social cohesion could be affected if families are dispersed across different projects, and the level of participation is the most limited among the alternatives, given that outcomes depend on market dynamics. This alternative does not guarantee that people can remain close to their traditional sources of income. That said, families would be integrated into mixed-income private developments, which may support urban integration and community participation in a new setting, which is why social acceptance in this case is medium.

Alternatives at a glance

Government Buyouts (with leaseback variant): The government purchases homes in high-risk areas, demolishes the structures, and restores the land. In the leaseback variant, families stay in their homes temporarily after the purchase, giving them more time to arrange permanent housing before demolition takes place.

Whole Community Relocation (WCR): The government builds a new community on a safer site within Esparza and relocates all households collectively, including infrastructure for schools, health services, and productive activities.

Land Swaps: The government exchanges public land in a safer location for the high-risk lots currently occupied by families. This removes households from the danger zone while providing them with a new lot to build on.

Transfer of Development Rights (TDRs): High-risk households receive development rights that they can sell to private developers. Developers use those rights to build at higher densities elsewhere in the municipality with tax exemptions, and in exchange, the relocated households receive a housing unit in the new development.

Cross-alternative analysis

Table 2 shows that differences between alternatives are not primarily determined by per-capita cost, which ranges between ₡14.1 and ₡17 million, but by three key dimensions: coverage, aggregate cost, and social sustainability.

Table 2. Outcomes Matrix

Alternative	Coverage (HH)	Timeline	Cost/HH (₡M)	Total cost (₡M)	Feasibility	Social acceptance
Direct buyouts	29	1 year	14.1	408	**High:** existing legal framework, few actors required, and precedents in Costa Rica	**Low:** fragments the community and doesn't cover the total cost of market values
Buyouts with leasebacks	29	5–7 years	15.5	449	**Low:** no existing framework for the government as landlord and no precedents	**Low:** fragments the community and doesn't cover the total cost of market values
WCR[1]	96	5 years	15.0	1,440	**Medium:** legal frameworks exist and there are precedents, but require complex	**High:** preserve community cohesion, secures access to

[1] The two best alternatives overall are highlighted in green.

Alternative	Coverage (HH)	Timeline	Cost/HH (₵M)	Total cost (₵M)	Feasibility	Social acceptance
					coordination with multiple stakeholders (communities and institutions)	livelihoods, and includes community participation
Land swaps	48	4 years	17.0	818	**Low:** would require legislative approval	**Medium:** it could provide access to livelihoods, but community cohesion is limited due to coverage constraints
TDRs	72[2]	3–4+ years	14.9	1,073	**Low:** no current legal frameworks and depends on market response	**Medium:** it could preserve some community ties, but participation is limited and it depends on market responses

[2] Potential coverage: 96 HH (100%). Estimated realistic coverage: ~72 HH (75%), subject to private market response.

Risk reduction. There is a clear divide between alternatives conditioned on property title (maximum coverage of 30 to 50%) and those that allow full inclusion (75 to 100%). WCR and TDRs are the only options that can reach the majority of the 96 families, though on different timelines: WCR offers certainty within five years, while TDRs depend on private market timing. The fastest alternatives (direct buyouts) cover only 30% of households, while those with higher coverage require more time and institutional coordination.

Costs. Aggregate costs reflect coverage differences: WCR and TDRs represent investments above ₡1,000 million, while buyouts and land swaps require between ₡449 and ₡818 million but only partially solve the problem. This creates a tradeoff between fiscal expenditure and effectiveness: investing more to cover 100% of households, or investing less and leaving 50 to 70% of the most vulnerable families without a solution. For WCR, the ₡1,440 million would be distributed over five years using international financing. For TDRs, the direct budget cost is lower since a portion corresponds to foregone tax revenues rather than direct expenditure.

Implementation feasibility. Only WCR reaches medium feasibility, as it can operate under existing legal frameworks (Law 8488), though it requires strong inter-institutional coordination. Direct buyouts are individually feasible, but the leaseback variant lacks a legal framework. Land swaps face the barrier of legislative approval, a lengthy and complex process. TDRs require Municipal Council approval and amendments to land use plans, both with high political uncertainty.

Social acceptance. This criterion is decisive. The San Martín case shows that relocations that do not preserve community cohesion and livelihoods have a high probability of failure (Calvo et al., 2024). WCR is the only alternative that combines high cohesion (collective relocation of the entire community), meaningful participation (FPIC, assemblies, relocation committee), and a design oriented toward livelihood preservation. Buyouts and land swaps fragment the community; TDRs offer urban integration to new communities but without a guarantee of collective cohesion.

Recommendation

Recommendation

The main strategic decision is between two models: WCR and TDRs. Both do not condition eligibility on property title, allowing them to reach between 75% and 100% of households (Polack et al., 2025).

The recommendation is that the CNE move forward with WCR as the primary alternative, given its greater institutional certainty, social legitimacy, and consistency with lessons learned.

Choosing WCR involves tradeoffs: it sacrifices the lower direct fiscal burden of TDRs and the speed of direct buyouts. However, buyouts can only reach 30% of households and fragment communities, the dynamic that caused the failure of San Martín (Calvo et al., 2024). The equity cost of excluding 70% of households outweighs the administrative advantages of speed. WCR also has the highest social acceptance: it relocates the community collectively, preserves cohesion, ensures participation through FPIC processes, and maintains proximity to livelihoods (Ajibade et al., 2022).

In terms of cost, WCR (₡15 million per household) is comparable to TDRs (₡14.9 million). The aggregate cost of ₡1,440 million is significant but includes comprehensive infrastructure, productive spaces, and livelihood support. TDRs represent a lower direct budget cost (approximately ₡868 million), since part of their costs are foregone tax revenues from exemptions. That said, WCR can be financed through international sources distributed over five years and can operate under Law 8488 without new legislation.

TDRs represent an innovative alternative with structural potential but greater uncertainty. They mobilize private investment and could reduce the direct fiscal burden. While there are no precedents in Costa Rica for disaster relocation, regional market conditions suggest favorable circumstances: cantons like Orotina and Esparza show high growth in new residential construction (CFIA, 2023). Where WCR offers greater public control and certainty, TDRs could open a new tool for future relocations in urban and coastal areas.

In conclusion, WCR should be the primary alternative due to its higher probability of effective and socially legitimate implementation. At the same time, the CNE should explore a TDR pilot framework with the municipality as a complementary and innovative tool for future risk-related relocations in Costa Rica.

> **Main takeaways**
>
> **Primary alternative:** WCR. It reaches all 96 households, has the highest social acceptance, operates under Law 8488 without new legislation, and can be financed through international sources over five years.
>
> **Complementary pilot:** TDRs. The CNE should explore a TDR pilot framework with the municipality as an innovative long term tool for future disaster-related relocations in Costa Rica.

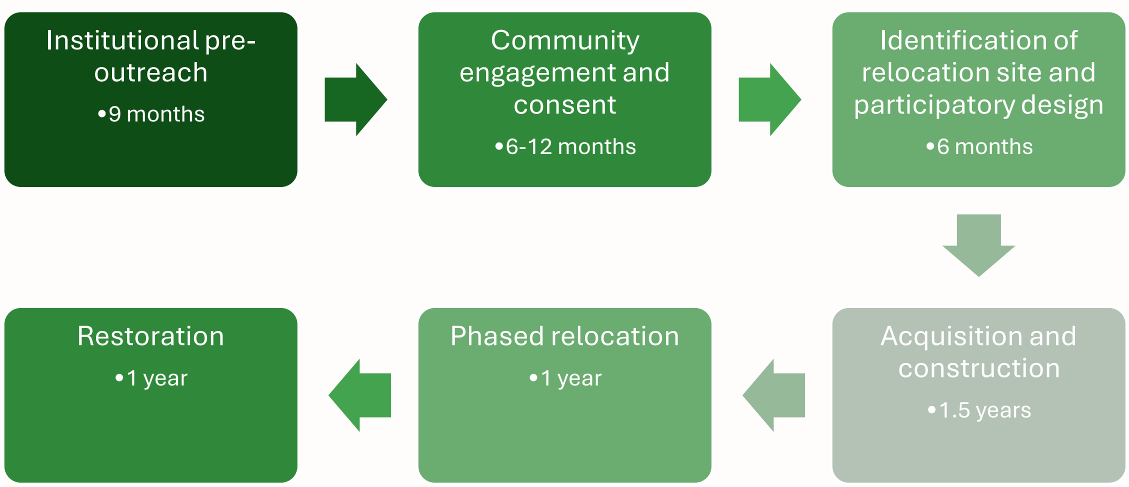
Implementation

Comparative experience and the failed case of San Martín in 2012 demonstrate that the success of a relocation depends critically on its implementation (Calvo et al., 2024; Polack et al., 2025). Getting the policy design right matters less if the institutional execution falls apart.

Implementation involves multiple stakeholders. The CNE coordinates and administers funds; the Municipality of Esparza acts as primary local executor and has shown active interest (E. Aguilar, personal communication, October 14, 2025). The Community Development Associations (ADCs) are strategic allies but their politicized nature also makes them a potential risk. MIVAH and BANHVI may present resistance due to jurisdictional disputes and incompatible BFV timelines. Other institutions (CFIA, Ministry of Education, Ministry of Health, IMAS) are structurally slow but generally neutral. This section details a six-phase plan (*Figure 3*) using a bottom-up approach that starts with institutional alignment before engaging communities.

Implementation model

Figure 3. Forward mapping of WCR implementation



Phase 1. Institutional pre-outreach and funds approval (9 months)

This first phase is essential because one important risk identified for the entire project is high institutional fragmentation and unclear institutional responsibilities around emergency housing. Before the CNE approaches 96 families with a relocation proposal, it needs the Municipality of Esparza, MIVAH, BANHVI, and other institutional partners at least provisionally committed. Opening a dialogue with communities without institutional alignment risks making promises the CNE cannot keep, which would burn trust that is very difficult to recover (Calvo et al., 2024).

The phase begins with a strategic negotiation between the CNE and the Municipality to confirm their willingness to co-lead. The Municipality has already shown active interest (E. Aguilar, personal communication, October 14, 2025), but this step formalizes roles (CNE as coordinator and fund administrator; Municipality as local executor for permits and logistics) through a formal cooperation agreement (convenio) that specifies obligations and accountability mechanisms.

Once the CNE-Municipality partnership is in place, the next step is broader institutional alignment. The strongest available tool is a presidential directive requiring all relevant institutions to cooperate and reduce administrative timelines. The CNE should also pursue agreements within the Sectoral Councils (Law 5525; Decree 38536-MP-PLAN), which create a public commitment that raises the political cost of non-cooperation, even if non-binding. Lastly, the CNE should negotiate bilateral agreements with strategic allies, such as the MIVAH and BANHVI, to increase the likelihood that they become champions of the project and help align political will in the broader institutional landscape.

In terms of financing, the primary financing mechanism is the Green Climate Fund (GCF), accessed through the Inter-American Development Bank as accredited entity and submitted jointly with the Ministry of Environment (MINAE). Application timelines typically range from 12 to 18 months from concept note to board approval (GCF, 2025), so the financing application must begin during this pre-outreach phase to have a provisional commitment in place before community engagement begins.

If GCF funding does not materialize in full, the contingency strategy has three layers: to issue a national emergency declaration to use the National Emergency Fund under Law 8488, which allows the CNE to access, receive and administer resources with reduced administrative burden; the Bono Familiar de Vivienda (BFV) administered by BANHVI, which can complement international funds for eligible families; and bilateral cooperation with development partners such as Central American Bank for Economic Integration (BCIE) or the Adaptation Fund, which have shorter application cycles than the GCF.

If total funding only reaches 60% of the ₡1,440M target, the project should be redesigned in phases rather than abandoned. The first construction phase would prioritize the families at highest risk, those in La Plaza and La Cueva where flood frequency is greatest, and defer the remaining households to a second phase contingent on additional funding. This phased approach preserves the project's viability, maintains institutional momentum, and avoids the political cost of presenting a proposal that collapses entirely if financing falls short. The CNE cannot present a ₡1.44 billion proposal to institutional partners without a credible financing plan that includes these contingencies.

The minimum commitments needed before community engagement begins are: (1) the Municipality confirms co-leadership; (2) MIVAH agrees to recognize the relocation project within its housing programs; (3) BANHVI commits to explore adapted BFV timelines or at minimum not block eligibility for participating families; and (4) the CNE secures a provisional financing commitment. Without these conditions, the CNE should not proceed to Phase 2.

Main risks and mitigation

Approval processes can take years without acceleration mechanisms. The potential resistance of BANHVI (incompatible BFV timelines) and MIVAH (jurisdictional disputes) aggravates this risk. The mitigation tools described above, from presidential directives to Sectoral Councils to bilateral agreements, form a fallback cascade from broadest to the most specific. In parallel, a national emergency declaration under Law 8488 allows the CNE to access the National Emergency Fund and expedite procurement under the General Public Procurement Law No. 9986.

Phase 2. Community Outreach and Consent (6 to 12 months)

The CNE and the Municipality will lead a dialogue process with the communities of La Plaza, La Cueva, and Caldera through working groups with the ADCs (relocation committee). From these dialogues, the ADCs will convene a community assembly (a power granted by their legal framework) to consult on the possible relocation and obtain the informed consent of families through individual consultations.

The most critical risk of the entire implementation comes from this phase: families may reject relocation if their basic needs are not addressed (employment, transportation, education, health), if their sense of belonging is threatened, or if they feel excluded from the process. The ADCs, given their politicized nature, may become active opponents and mobilize the community against the project, the worst-case scenario. The mitigation strategy consists of formally involving the ADCs from the outset and assigning them a role in the project, shifting the locus of responsibility toward them. This transforms them from passive recipients into active participants, increasing their potential gains and losses based on the project's success or failure, and generating incentives for their cooperation.

A second potential risk is getting only partial consent. The managed retreat literature shows that full consent is rare and that holdouts can stall entire projects (Siders, 2019; Calvo et al., 2024). The CNE needs a clear threshold: the project should proceed if at least 75% of households (approximately 72 of 96 families) give informed consent. This threshold balances two concerns. Below it, the project lacks the critical mass to justify the investment in shared infrastructure, and the political legitimacy of the relocation weakens. Above it, requiring unanimity gives any small group of holdouts effective veto power over the entire project.

Families who decline would remain eligible for existing risk management mechanisms: emergency shelter, temporary housing assistance through the CNE, and social support through IMAS. They would not be forcibly evicted. The practical implication for project design is that the new site should be sized for 75% participation initially, with flexibility to accommodate additional families in a later phase if holdouts change their minds.

The last potential risk is finding contrasting positions between ADCs. The ADC co-optation strategy assumes the three communities' ADCs will act as a unified actor, but in practice they may have different priorities on site selection and housing design. To mitigate this, the CNE should establish clear decision rules before the site selection process begins, including a weighted scoring system based on objective criteria that all three ADCs agree to in advance. If consensus cannot be reached, the CNE retains final decision authority as project coordinator, though this carries political costs and should be treated as a last resort.

Main risks and mitigation

Families can be resistant to relocation if their needs are not addressed. This could lead to a low take-up or even the project's failure. Mitigation strategies to address this concern include the inclusion of ADCs as a key stakeholder, addressing partial take-up, and looking for mutually consensual rules to resolve disputes between communities' preferences.

Phase 3. Site Identification and Participatory Design (6 months)

A permanent working group will be established with the ADCs to select the relocation site and design the urban project according to community needs. Calvo et al. (2024) and IOM (2022) identify the lack of community participation as a key factor in the failure of previous relocations. In San Martín, residents cited disconnection from their livelihoods and the absence of participation as reasons for rejection. This phase therefore seeks to return responsibility to communities so they take ownership of the project, reducing the risk of rejection identified in the previous phase. The institutional risks of this phase are equivalent to those of Phase 2, and the same mitigation measures apply.

Phase 4. Land Acquisition and Construction (2 years)

This phase is the most institutionally complex. It requires construction and development permits (CFIA, Municipality), insurance coordination, procurement of private developers through public tender, and BFV disbursements from BANHVI for eligible families. The remainder will be financed through the National Emergency Fund (described in Phase 1).

With strategic alignment secured in Phase 1, on-the-ground coordination during construction requires activating the Municipal Emergency Committees, which bring all institutions with local presence together to coordinate logistics and project execution. The Municipality chairs this space with the CNE co-leading. This is the operational layer that translates high-level institutional commitments into day-to-day coordination: scheduling inspections, sequencing BFV disbursements with construction milestones, and resolving bottlenecks when multiple institutions must act in sequence.

Through these committees, the CNE should look to show early wins: political acts of inauguration and visible progress in construction to increase the stakes in the project.

Main risks and mitigation

The main risk in this phase is that the institutional coalition assembled in Phase 1 erodes during a long construction process. Three mechanisms serve that function: the ADC co-optation strategy ties community political capital to the project's success; committed international financing creates external accountability, since abandoning a GCF-funded project carries reputational costs; and creating early wins by making visible construction progress generates momentum that makes continuation politically easier than cancellation. The CNE should manage these dynamics deliberately by publicizing milestones and ensuring each partner's contribution is recognized.

Phase 5. Phased Relocation (1 year)

Once housing is built, the CNE will coordinate the phased relocation. The Municipality will manage logistics, maintain a registry of relocated families, and disburse monetary support. The Ministry of Education will relocate the local school; the Ministry of Health will relocate the primary care clinic; and IMAS will provide social accompaniment and rights protection during the transition. The cooperation of these institutions will be backed by the presidential directive activated in Phase 1. The institutional risks of this phase are equivalent to those of Phase 4, and the same mitigation measures apply.

Phase 6. Restoration (1 year)

The Municipality will demolish the remaining structures at the original site and build parks and public plazas to prevent reoccupation by new informal settlements.

Cross-cutting risk: changes in political priorities

Implementation spans from 5 to 5.5 years, which means that at least one change of government is likely (presidential and municipal terms are four years). A new administration could deprioritize the project. The bottom-up strategy (Weimer & Vining, 2017) addresses this risk: community and ADC ownership of the project creates social pressure that raises the political cost of cancellation. Committed international financing from early phases also reduces vulnerability to internal budgetary changes.

Conclusion

The communities of La Plaza, La Cueva, and Caldera in the canton of Esparza face a level of exposure to flooding and coastal erosion that prevention and mitigation alone cannot address. Ninety-six households need a permanent relocation solution, and the existing institutional framework, built around temporary emergency assistance and individual low-income housing subsidies, is not designed to deliver one. The failed attempt to relocate families from San Martín in 2012 shows what happens when relocation is treated as a housing logistics problem rather than a community development challenge.

Whole Community Relocation is the recommended alternative for this case because it is the only option that reaches all affected households, preserves community cohesion, and incorporates the participatory mechanisms that the evidence identifies as essential for success. The six-phase implementation plan prioritizes institutional alignment before community engagement, establishes a clear consent threshold, and includes a credible financing strategy with contingencies. If implemented as designed, this project would not only resolve the immediate displacement crisis in Esparza but also establish a replicable model for planned relocation in Costa Rica, a country that currently lacks one. The CNE has the legal authority, the institutional relationships, and the international cooperation channels to lead this effort.

However, the Esparza case also reveals a deeper problem that goes beyond these three communities. Climate change will continue to increase the frequency and scale of displacement events across the country. No single solution can address every situation, and Whole Community Relocation, while effective when an entire community is affected, is expensive and takes a long time to implement. It cannot be the default response to every emergency housing need.

One of the main findings of this research is that Costa Rica needs a sustained and permanent emergency housing model that includes different response mechanisms depending on the needs of each family and community. This requires a review of institutional roles and clearer coordination between CNE, the various housing institutions, and local governments. It also means revisiting how funding sources work and how resources reach the people who need them. The current BFV model is long, complex, and places a heavy administrative burden on families. If this is not resolved, it will continue to create serious problems over time, not only for people displaced during emergencies but for all potential beneficiaries.

This work should be seen as a portfolio of policy alternatives rather than a single recommendation. In some situations, when families are displaced locally, a buyout may be the best response. When an entire community is affected, WCR may be the right fit. In others, where real estate development is stronger, TDRs may be more appropriate. Each model has its strengths, trade-offs, and weaknesses, and each is designed to solve a different kind of problem.

The most disruptive of these mechanisms for the Costa Rican context is TDRs. It breaks away from the country's traditional state-centered approaches and focuses on building public-private partnerships to solve a public problem with the least possible deadweight loss. As climate change continues to drive more frequent emergencies and larger displaced populations, mechanisms like TDRs deserve serious attention given that needs are always unlimited but public budgets are not.

Based on this document, CNE now has a concrete set of alternatives to define one or more emergency housing mechanisms in practice, and to choose from based on the specific needs of each community.

Annex

Annex 1. APP project matrix and financial calculations

APP project matrix.xlsx

References

Abu, M., Heath, S. C., Adger, W. N., Codjoe, S. N. A., Butler, C., & Quinn, T. (2024). Social consequences of planned relocation in response to sea level rise: Impacts on anxiety, well-being, and perceived safety. *Scientific Reports*, *14*(1), 3461. https://doi.org/10.1038/s41598-024-53277-9

Ajibade, I., Sullivan, M., Lower, C., Yarina, L., & Reilly, A. (2022). Are managed retreat programs successful and just? A global mapping of success typologies, justice dimensions, and trade-offs. *Global Environmental Change*, *76*, 102576. https://doi.org/10.1016/j.gloenvcha.2022.102576

Allen, B. (2020). *SB 1293- AMENDED*. https://leginfo.legislature.ca.gov/faces/billTextClient.xhtml?bill_id=201920200SB1293

Asamblea Legislativa. (1949). *Constitución Política de Costa Rica*. https://pgrweb.go.cr/scij/busqueda/normativa/normas/nrm_texto_completo.aspx?nValor1=1&nValor2=871

Asamblea Legislativa. (1962). *Ley de Tierras y Colonizacion*. https://pgrweb.go.cr/SCIJ/Busqueda/Normativa/Normas/nrm_texto_completo.aspx?param1=NRTC&nValor1=1&nValor2=32840&nValor3=145033&strTipM=TC

Balachandran, B., Olshansky, R. B., & Johnson, L. A. (2022). Planning for Disaster-Induced Relocation of Communities. *Journal of the American Planning Association*, *88*(3), 288–304. https://doi.org/10.1080/01944363.2021.1978855

Banco Hipotecario de la Vivienda [BANHVI]. (n.d.). *Extrema Necesidad*. Retrieved November 13, 2025, from https://www.banhvi.fi.cr/bono/programas_especiales/extrema_necesidad.aspx

Banco Hipotecario de la Vivienda [BANHVI]. (2025). *El Bono Familiar de Vivienda*. https://www.banhvi.fi.cr/publicaciones/folletos/FolletoBFV.pdf

Binder, S. B., & Greer, A. (2016). The Devil Is in the Details: Linking Home Buyout Policy, Practice, and Experience After Hurricane Sandy | Article | Politics and Governance. *Policits and Governance*, *4*(4). https://www.cogitatiopress.com/politicsandgovernance/article/view/738

Bower, E. R., Epifantseva, S., Schmitter, S., Wong-Parodi, G., Kulp, S., & Field, C. B. (2024). Planned relocation may reduce communities' future exposure to coastal inundation but effect varies with emission scenario and geography. *Communications Earth & Environment*, *5*(1), 670. https://doi.org/10.1038/s43247-024-01854-1

Brookings Institution. (2015). *Guidance on Planned Relocation.pdf | Powered by Box*. https://georgetown.app.box.com/s/qwx6dcvl9762fv9itnqn98ogx1h3sjzz

Calvo, M. P., Cordero, M. A., & Robles, A. (2024). *Estudio de Diagnóstico de la Movilidad por Cambio Climático en la Comunidad de Caldera, Esparza, Puntarenas | OIM Costa Rica*. Organización Internacional para las Migraciones OIM. https://costarica.iom.int/es/resources/estudio-de-diagnostico-de-la-movilidad-por-cambio-climatico-en-la-comunidad-de-caldera-esparza-puntarenas

Centeno-Morales, J., Vargas-Trejos, Y., González-Varela, M., & Alfaro-Orozco, E. (2020). Case study: Lajas Compartir, a rapprochement to environmental displacement in Costa Rica. *Revista Geográfica de América Central*, (64), 158–180. https://doi.org/10.15359/rgac.64-1.7

Colegio Federado de Ingenieros Y Arquitectos [CFIA]. (2023). *Balances y tendencias del sector de vivienda 2023*. https://cfia.or.cr/site/wp-content/uploads/2024/pdf/descargas/informes/balance-y-tendencias-sector-vivienda.pdf

Comisión Nacional de Prevención de Riesgos y Atención de Emergencias [CNE]. (2025). *Tablero General de Daños y Perdidas*. https://cnecr.maps.arcgis.com/apps/dashboards/618974dd00304c429b89f395b5386f59

Comisión Nacional de Prevención de Riesgos y Atención de Emergencias [CNE]. (2026). *CNE*. https://www.cne.go.cr/acerca/historia.aspx

Fernández Arce, M., Ramírez Umaña, C., & Muñoz Jiménez, R. (2020). Prevención del riesgo en Santo Domingo de Heredia, Costa Rica: Reubicación del barrio Fátima. *Revista de Estudios Latinoamericanos sobre Reducción del Riesgo de Desastres (REDER)*, *4*(2), 126–133.

Georgetown Climate Center [GCC]. (2020). *Managed Retreat Toolkit » Transfer of Development Rights—Georgetown Climate Center*. Georgetownclimatecenter.Org.

https://www.georgetownclimate.org/adaptation/toolkits/managed-retreat-toolkit/transfer-of-development-rights.html

Green Climate Fund [GCF]. (2025, April 10). *Access funding* [Text]. Green Climate Fund. Green Climate Fund. https://www.greenclimate.fund/projects/access-funding

Hoang, T., & Noy, I. (2023). The income consequences of a managed retreat. *Regional Science and Urban Economics*, *100*, 103896. https://doi.org/10.1016/j.regsciurbeco.2023.103896

Instituto Nacional de Vivienda y Urbanismo [INVU]. (2025a). *Instituto Nacional de Vivienda y Urbanismo*. https://www.invu.go.cr/

Instituto Nacional de Vivienda y Urbanismo [INVU]. (2025b). *Proyectos INVU*. https://www.invu.go.cr/

Intergovermental Panel on Climate Change [IPCC]. (2014). *Changements climatiques 2014: Incidences, adaptation et vulnérabilité : résumés, foire aux questions et encarts thématiques : contribution du groupe de travail II au cinquième Rapport d'évaluation du Groupe d'experts intergouvernemental sur l'évolution du climat*. Organisation météorologique mondiale.

Intergovernmental Panel On Climate Change (IPCC)). (2023). *Climate Change 2022 – Impacts, Adaptation and Vulnerability: Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (1st ed.). Cambridge University Press. https://doi.org/10.1017/9781009325844

Keeler, A. G., Mullin, M., McNamara, D. E., & Smith, M. D. (2022). Buyouts with rentbacks: A policy proposal for managing coastal retreat. *Journal of Environmental Studies and Sciences*, *12*(3), 646–651. https://doi.org/10.1007/s13412-022-00762-0

Koslov, L. (2016). The Case for Retreat. *Public Culture*, *28*(2 (79)), 359–387. https://doi.org/10.1215/08992363-3427487

La Nación, P. R. de L. (2014, July 5). *Altas olas inundan 70 viviendas en Caldera*. La Nación. https://www.nacion.com/sucesos/desastres/altas-olas-inundan-70-viviendas-en-caldera/EZZTOSBPLFGKTMSDHBIRD7ESMI/story/

Laska, S. (Ed.). (2020). *Louisiana's Response to Extreme Weather: A Coastal State's Adaptation Challenges and Successes*. Springer International Publishing. https://doi.org/10.1007/978-3-030-27205-0

Lineamientos Para La Definición de Territorios y Población Objetivo Para El Desarrollo de Vivienda Individual y Proyectos Habitacionales Financiados Con Recursos Del Sistema Financiero Nacional Para La Vivienda, Nos. 43713-MP-MIVAH (2022). https://pgrweb.go.cr/scij/Busqueda/Normativa/Normas/nrm_texto_completo.aspx?param1=NRTC&nValor1=1&nValor2=51456&nValor3=133563&strTipM=TC

McAdam, J., & Ferris, E. (2015). Planned Relocations in the Context of ClimateChange: Unpacking the Legal and Conceptual Issues. *Cambridge Journal of International and Comparative Law*, *4*(1), 137> – 166. https://doi.org/10.7574/cjicl.04.01.137

Miao, Q., & Davlasheridze, M. (2022). Managed Retreat in the Face of Climate Change: Examining Factors Influencing Buyouts of Floodplain Properties. *Natural Hazards Review*, *23*(1), 04021063. https://doi.org/10.1061/(ASCE)NH.1527-6996.0000534

Ministerio de Ambiente y Energía [MINAE]. (2018). *Política Nacional de Adaptación al Cambio Climático de Costa Rica 2018-2030*. Gobierno de Costa Rica. https://cambioclimatico.minae.go.cr/wp-content/uploads/2019/01/Politica-Nacional-de-Adaptacion-al-Cambio-Climatico-Costa-Rica-2018-2030.pdf

Ministerio de Ambiente y Energía [MINAE]. (2022). *Plan de Acción Regional para la Adaptación al Cambio Climático 2022-2026*. Gobierno de Costa Rica. https://cambioclimatico.minae.go.cr/wp-content/uploads/2022/05/Plan-de-Accion-ACC_Pacifico-Central_.pdf

Ministerio de Hacienda. (2017). *Mapa de Valores de Terrenos por Zonas Homogéneas Orotina*. https://www.hacienda.go.cr/docs/2017_209_PVTZH.pdf

Ministerio de Hacienda. (2018). *Mapa de Valores de Terrenos por Zonas Homogéneas Puntarenas*. https://www.hacienda.go.cr/docs/2018_604_PVTZH.pdf

Ministerio de Hacienda. (2019). *Mapa de Valores de Terrenos por Zonas Homogéneas Esparza*. https://www.hacienda.go.cr/docs/602_Esparza_Cantonal.pdf

Ministerio de Hacienda. (2023). *Manual de Valores Base Unitarios por Tipología Constructiva*. Diario La Gaceta.

https://www.hacienda.go.cr/docs/ManualdeValoresBaseUnitariosporTipologiaConstructiva.pdf?utm_source

Municipalidad de Esparza. (2025a). *Escala de categoría, salario base y anualidad 2025 · Datos Abiertos de la Municipalidad de Esparza*. MunicipalidaddeEsparza. https://datosabiertos.muniesparza.go.cr/dashboards/21812/escala-de-categoria-salario-base-y-anualidad-2025/

Municipalidad de Esparza. (2025b). *Plan Cantonal de Emergencia de Esparza*. Comité Municipal de Emergencia Esparza.

Navas, J. (2025). *Diagnóstico de la Vivienda de Emergencia Costa Rica*. Programa de Emergencia para la Reconstrucción Integral y Resiliente de Infraestructura.

Organización Internacional para las Migraciones (OIM). (2022). *Movilidad Humana en contextos de desastres, degradación ambiental y cambio climático en Costa Rica: Análisis de evidencia y de políticas*. https://costarica.iom.int/sites/g/files/tmzbdl1016/files/documents/Policy%20Brief%20Series_Costa%20Rica_WEB.pdf

Petz, D. (2015). PLANNED RELOCATIONS IN THE CONTEXT OF NATURAL DISASTERS AND CLIMATE CHANGE: A REVIEW OF THE LITERATURE. *Climate Change*.

Pérez, R., & Zelmeister, I. (2011). Respuesta a inundaciones recurrentes: Programa de autoconstrucción asistida de viviendas. In *Reasentamiento preventivo de poblaciones en riesgo de desastre. Experiencias de América Latina*. The World Bank. https://documents1.worldbank.org/curated/en/564081468053039674/pdf/702830ESW0P1100ventive0Resettlement.pdf

Pinter, N. (2021). The lost history of managed retreat and community relocation in the United States. *Elementa: Science of the Anthropocene*, *9*(1), 00036. https://doi.org/10.1525/elementa.2021.00036

Polack, D., Gamell, P., Escribano, P., Fernández, C., & Calvo Barboza, P. (2025). *Herramienta de Costeo para Fondos (HCF) sobre perdidas y danos en contexto de movilidad humana en América Latina y el Caribe, C - Costa Rica*. , OIM - La Ruta del Clima. https://lac.iom.int/sites/g/files/tmzbdl2601/files/documents/2025-10/oim_hcf_estudio_de_caso_costa_rica_1.pdf

Programa de las Naciones Unidas para el Desarrollo [PNUD]. (2024). *ATLAS DE DESARROLLO HUMANO CANTONAL EN COSTA RICA 2024*. https://pnud-conocimiento.cr/wp-content/uploads/2024/10/atlas-cantonal-reporte-2.pdf

Sandoval, V., Voss, M., Flörchinger, V., Lorenz, S., & Jafari, P. (2023). Integrated Disaster Risk Management (IDRM): Elements to Advance Its Study and Assessment. *International Journal of Disaster Risk Science*, *14*(3), 343–356. https://doi.org/10.1007/s13753-023-00490-1

Siders, A. R. (2019). Social justice implications of US managed retreat buyout programs. *Climatic Change*, *152*(2), 239–257. https://doi.org/10.1007/s10584-018-2272-5

Stern, S. M. (2022). Climate Transition Relief: Federal Buyouts for Underwater Homes. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.4037682

Tronquet, C. (2015). From Vunidogoloa to Kenani: An Insight into Successful Relocation. *The State of Environmental Migration*. http://labos.ulg.ac.be/hugo/wp-content/uploads/sites/38/2017/11/The-State-of-Environmental-Migration-2015-121-142.pdf

UN General Assembly (UNGA). (2016). *A/71/644. Report of the open-ended intergovernmental expert working group on indicators and terminology relating to disaster risk reduction*. United Nations. https://www.preventionweb.net/files/50683_oiewgreportenglish.pdf

United Nations. (2015). *Marco de Sendai para la Reducción del Riesgo de Desastres 2015-2030*.

UN High Commissioner for Refugees [UNHCR]. (2017). *A Toolbox: Planning Relocations to Protect People from Disasters and Environmental Change*. Refworld. https://www.refworld.org/policy/opguidance/unhcr/2017/en/118118

Weimer, D. L., & Vining, A. R. (2017). *Policy Analysis: Concepts and Practice* (6th ed.). Routledge. https://doi.org/10.4324/9781315442129

White, L. (2022). *Managed Retreat: An Introduction and Exploration of Policy Options*. American Meteorological Society. https://doi.org/10.1175/managed-retreat-2022

World Bank. (2018). *Regional Workshop on Planned Relocations to protect persons from disasters and environmental change in the Latin American context*. Global

Knowledge Partnership on Migration and Development KNOMAD. https://knomad-ois.assyst.in/sites/default/files/2018-07/KNOMAD%20WORKSHOP_PLANNE%20RELOCATIONS_SYNTHESIS%20NOTE_MAY%202018.pdf

Yara, S. (2025, March 5). ¿Cuánto cuesta un avalúo en Costa Rica? Factores y Precios. *Tinsa Costa Rica*. https://www.tinsa.cr/costo-avaluo-costa-rica/